College of Engineering and Technology
Department of Computer Science and Engineering

Assignment of Fall Semester 2025

Program: Bachelor of Computer Science and Engineering

Course Instructor: Kashfi Shormita Kushal

Course Code: CSC 451
Course Title: Management Information System **Full Marks: 100**
Section: A **Submission Date: 07-01-2026**

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Analysis and Risk Mitigation of the Bangladesh Bank Heist

In 2016, the Bangladesh Bank was breached, resulting in the loss of $81 million. This highly sophisticated attack is seen by many as the largest bank robbery in history, later found to be the works of nation state hackers.

The attack was meticulously calculated, beginning a year before the eventual release of such large funds. It is suspected that emails containing malware were sent to employees of the Bangladesh Bank- seemingly harmless files which, when opened, released malware onto the user's computer. From here, the attackers were able to gain access to the wider systems within the Bangladesh Bank. In the meantime, bank accounts were being set up across the world for the eventual transfer of money.

The attackers were also able to hack into the *SWIFT* network through the Bangladesh Bank. The *SWIFT* network is used to transfer money internationally between banks, and the attackers were knowledgeable of the system, suggesting they had carried out attacks like this before. Having access to the *SWIFT* network meant that they could transfer money out of the Bangladesh Bank's account in the New York Federal Reserve.

The plan was extremely complex, and took advantage of the different time zones involved in international banking. Transferring money from the New York account of the Bangladesh Bank into bank accounts in the Philippines meant the involvement of three time zones. The timing was expertly planned, meaning that when the New York Fed began receiving fraudulent payment orders on Thursday afternoon, the Bangladesh Bank was closed (their weekend begins on Friday). Then, when the Bangladesh bank reopened on the Sunday, the New York Fed was of course closed for the weekend. At this realization, the Bangladesh Bank tried to contact the banks in the Philippines, which was closed for Chinese New Year. The attack came at a time

where banks were unable to communicate effectively, increasing their chance of success. By the time the banks in their respective countries were able to contact each other, it was too late.

A total of almost $1 billion was ordered to be transferred, though only $81 million was ever actually transferred, as many of the requests were rejected by the New York Fed. However, $81 million is still an extremely vast sum of money. A percentage of the stolen money was immediately sent to a Chinese national, though it is unclear why- perhaps this was someone who had played a role in the attack. The rest of the money, which was sent to accounts across the world, including the Philippines and Sri Lanka, needed to be laundered. This was achieved by sending the money to two casinos in the Philippines, where the hackers gambled and cashed out, leaving no trace of the money.

This attack was extremely advanced, stealing such a huge amount of money and involving national banks across the world. It is now believed that North Korean nation state hackers are responsible for the heist. The operation fit the patterns of the workings of the *Lazarus Group*, thought to be associated with North Korea. North Korea is reported to employ over 1,500 hackers across the world, supported by around 5,000 other staff. North Korean hackers are known for their skill in targeting banks in developing nations, especially in Southeast Asia, who may have more security vulnerabilities than countries long-established in the market. The traits of the *Lazarus Group* found throughout the heist, combined with the known abilities of North Korean hackers in infiltrating banks newly emerging into the market, leads to the strong likelihood that North Korean nation state hackers launched the attack.

It is, however, very unexpected to see nation state hackers attack for financial gain. Though, in this case, it does seem to make sense. In 2013, sanctions were passed to restrict North Korea's ability to transfer large sums of money internationally, due to their launching of missile tests against the wishes of other countries. Therefore, this attack may be seen as a response to this sanction- taking matters into their own hands by stealing from international banks.

This example of nation state hacking demonstrates the immense expertise, forethought and co-ordination of these groups and their attacks. Such large amounts of money stolen, likely never to be retrieved, and the involvement of countries on the international scale highlights just how competent nation state attackers can be, and how their attacks can be devastating.

Term Paper Topic:

Analysis and Risk Mitigation of the Bangladesh Bank Heist

1. Task:

In a detailed report, address the following:

1. **Problem Identification and Analysis:**
    - Analyze the Bangladesh Bank Heist case study to identify the vulnerabilities exploited by the attackers.
    - Describe the technical details of the Bangladesh Bank Heist, including the role of malware, and SWIFT network exploitation.
    - Discuss the impact of cultural, organizational, and time-zone factors in the success of the attack.
2. **System Requirements and Constraints**
    - Outline the essential technical and managerial requirements for a secure banking information system.
    - Identify constraints, including technical, operational, and regulatory, that could affect the design and implementation of a robust security system.
3. **Risk Assessment and Mitigation**:
    - Evaluate potential risks for similar financial institutions using information from Management Information Systems (MIS) principles.
    - Propose mitigation strategies, focusing on both technical solutions and managerial practices.
4. **System Design Proposal:**
    - Design a high-level architecture for secure financial information system institutions like Bangladesh Bank.
    - Address how these solutions comply with international security standards and legal requirements.
    - Include diagrams, models, or frameworks where applicable to support your recommendations.
5. **Ethical and Social Considerations:**
    - Assess the ethical responsibilities of engineers and managers in securing information systems.
    - Analyze the impact of such breaches on societal trust, economic stability, and international relations.
    - Highlight how the proposed solutions align with societal, health, safety, legal, and cultural considerations.

Justification on PO(f) (The Engineer and Society) Attainment:

Assignment Features	**Knowledge Profile & Attributes of Complex Engineering Problem**	
Students apply MIS theories, cybersecurity principles, and SWIFT network operations to analyze vulnerabilities and propose technical solutions. This involves applying bodies of knowledge that are at the forefront of financial and information systems security.	**K4**	**P1- Depth of Knowledge**
The case requires understanding the societal, legal, and cultural implications of cyberattacks on national banks. Students must assess ethical responsibilities, public trust, and international relations—key roles of engineers in society.	**K7**	
The problem demands balancing conflicting needs—security vs. usability, national vs. international legal standards, and privacy vs. transparency—when designing a secure financial system.	**P2- Range of Conflicting Requirements**	
The assignment demands high-level analytical thinking: dissecting technical, managerial, and geopolitical causes of the heist, assessing failures, and crafting resilient countermeasures based on real-world data.	**P3- Depth of Analysis Required**	
Multiple stakeholders are involved: Bangladesh Bank, SWIFT network, Federal Reserve, casinos, governments, and the public. Students must consider their conflicting needs while proposing systemic, ethical, and sustainable solutions.	**P6- Extent of stakeholder involvement and level of conflicting requirements**	

Report Guidelines

Title Page

- Title: *Analysis and Risk Mitigation of the Bangladesh Bank Heist*
- Author's Name
- Institution
- Course Title
- Submission Date

Executive Summary

- Brief overview of the Bangladesh Bank Heist.
- Summary of identified vulnerabilities, impacts, proposed solutions, and ethical considerations.

Table of Contents

- Introduction
- Problem Identification and Analysis
- System Requirements and Constraints
- Risk Assessment and Mitigation
- System Design Proposal
- Ethical and Social Considerations
- Conclusion
- References

1. Introduction

- Background of the Bangladesh Bank Heist (based on the case study provided).
- Importance of securing financial information systems.
- Objectives and scope of the report.

2. Problem Identification and Analysis

- **Technical Details**: Role of malware, SWIFT network exploitation, and time-zone manipulation.
- **Organizational Vulnerabilities**: Cultural and procedural gaps contributing to the attack.
- **Impact**: Financial, reputational, and systemic effects.

3. System Requirements and Constraints

- **Technical Requirements**: Secure communication protocols, malware defenses, and system monitoring.
- **Managerial Requirements**: Policies for access control and employee training.
- **Constraints**: Regulatory, operational, and cost considerations.

4. Risk Assessment and Mitigation

- **Risk Identification**: Key threats to financial systems based on MIS principles.
- **Mitigation Strategies**:
  - **Technical**: Multi-layered security architecture, frequent audits, and real-time threat detection.
  - **Managerial**: Enhanced training, cross-time-zone communication protocols, and accountability systems.

5. System Design Proposal

- **High-Level Architecture**: Diagram and explanation of a secure financial information system.

- **Compliance**: Alignment with international standards like PCI DSS and ISO 27001.
- **Implementation Details**: Role of encryption, firewalls, intrusion detection systems, and robust SWIFT integration.

6. Ethical and Social Considerations

- **Ethical Responsibilities**: Role of engineers and managers in preventing security lapses.
- **Societal Impact**: Loss of public trust and economic stability.
- **Alignment with Values**: How proposed measures respect cultural and legal norms.

7. Conclusion

- Recap of findings and solutions.
- Emphasis on the need for continual improvements in cybersecurity.
- Call to action for financial institutions to prioritize security.

8. References

- Use APA citation style for all sources, including the uploaded documents and additional research if applicable.

Instructions:

1. This is a group task.
2. The full report must be computer-composed and formatted neatly with proper margins.
3. Each group must submit:
    - One hard copy of the spiral-bound report, signed by all group members.
    - A soft copy of the report to the **google classroom post.**
4. The report carries **50% marks and viva 50% marks** (Out of 100).
5. A group member contribution table must be attached to the report.
6. The last date for submission is **7th January 2026**.